Rare-event threshold estimation: a critical-margin upper bound, a stopping obstruction, and an audited research checkpoint

AI-assisted research in the adaptive-threshold-estimation repository

27 October 2026

Abstract

The arbitrary-law rare-event minimax conjecture remains unresolved in this checkpoint. A new paper proof supplies an unbiased observable estimator under a quadratic margin envelope with second moment at most $165\rho/56$ and mean work at most $1350a^{-2}L(1+\log L)$, where $L = 1+\log(1/\rho)$. Its budgeted risk is within a factor of order $\sqrt{1+\log L}$ of the audited candidate lower bound for the exact quadratic law. We neither remove this factor nor prove it necessary. A counterexample shows that one-law finite expected stopping time cannot transfer almost-sure termination to the other law in an unbounded fresh-coin reservoir. A corrected finite-prefix testing principle explains the role of uniform budgets. An elementary qualified unknown-rarity guarantee and an arbitrary-law finite upper bracket are also given. New direct-observation experiments have mixed and sometimes poor performance; no practical superiority is established. Primary-source comparison identifies the exact scope of the inherited nested-simulation lower bound and important model exclusions. All new statistical theorems are paper proofs, not Lean proofs; novelty and external specialist review remain unresolved. The v4 paper and frozen baselines are preserved.

1 Experiment, computation, and evidence contract

A fresh opaque coin independently draws $D \sim \mu$ on $(0, 1/2]$ and $S \mid D = d \sim \text{Bernoulli}(q(d))$. Its retained probability is $P = 1/2 + (2S-1)D$. Observations of that coin are conditionally iid Bernoulli(P). The algorithm observes only bits and identities. It may create, revisit, interleave, abandon, and stop. Every requested observation costs one; independent randomization and finite deterministic computation are uncharged. Creating unused coins gives no information and can be deferred to first use. No magnitude-selected query, sign oracle, or oracle for q is supplied.

For known $0 < \rho \le 1/4$, define

$$\mathcal{Q}_\rho(\mu) = \{q \text{ measurable} : 0 \le q \le 1,\ \int q\, d\mu \le \rho\}, \qquad \theta = \int q\, d\mu.$$

This is a global constraint; $q(d)$ may equal one on a small region. $R_{\mu,\rho}(B)$ is the infimum of worst-case RMS error over algorithms with $\sup_{q\in\mathcal{Q}_\rho} \mathbb{E}_q T \le B$. $r_{\mu,\rho}(B,\delta)$ is the infimum radius for which an algorithm under the same uniform expected-work constraint has $\sup_q \Pr_q(|\widehat{\theta} - \theta| > r) \le \delta$. Algorithms and their design constants may depend on known μ, ρ, B, δ. The new critical-envelope algorithm uses only a, ρ, B, δ, not μ.

For arbitrary known laws, the finite upper bracket below permits precomputed real integral constants or certified upper bounds. This is the mathematical known-law minimax convention, not a universal computability theorem for every representation of a measure. A sampled coin's magnitude remains hidden. Infinite arithmetic operations between two observations are not allowed. An explicit experiment exists on a countable product of independent magnitude

variables, uniform variables producing signs, and uniform flip tapes. All algorithms below make finite operations at each step and have the stated termination property. Their random seed is independent of this product.

Pinned baseline. Initial HEAD was `ddf950122776a059e62c808021bbd2b59c39f0b0`, with preexisting uncommitted route-a research and documentation. The new local branch is `research/rare-minimax-20261027`. Concurrent activity later committed that inherited work and edited separate Tier 5 files; source hashes and the precise audit scope are recorded in `STATE.md` and `FORMAL_COVERAGE.md`. No concurrent work is credited as this run's construction. No remote publication or researcher contact was performed.

Evidence labels. FORMALIZED means only an exact checked Lean declaration with its assumptions and audited dependency closure. PROVED-ON-PAPER means the argument is written, with explicit dependencies; agent review is not external peer review. CONJECTURE, NUMERICAL, DISPROVED, and PRIOR-ART / UNRESOLVED-PRIORITY are used literally. Exact rational finite checks certify those calculations only. The new results below have no corresponding complete Lean statistical theorem.

2 The candidate, inherited results, and what remains open

The proposed functional is

$$\Psi_{\mu,\rho}(h) = \sup_{0\le w\le 1,\, \mathbb{E}_\mu w\le \rho} \mathbb{E}_\mu \min\{w(D), h\sqrt{w(D)}/D\}.$$

The existing arbitrary-law optimizer is $w_*(d) = \min\{1, t^2/d^2\}$, where $\mathbb{E}\min(1, t^2/D^2) = \rho$. This variational statement is FORMALIZED in `ThresholdHC.Tier3.arbitrary_law_optimizer`; it does not supply a statistical algorithm. The cutoff exists uniquely for $0 < \rho < 1$: dominated convergence gives continuity from zero to one, and strict increase holds before saturation because there is positive mass above the smaller cutoff. Atoms require no allocation of selected magnitudes. For $h < t$ the value is

$$\mu(D \le h) + h\mathbb{E}[D^{-1}\mathbf{1}_{\{h<D\le t\}}] + ht\mathbb{E}[D^{-2}\mathbf{1}_{\{D>t\}}],$$

and for $h \ge t$ it is ρ. The pieces agree at boundary atoms. For a fixed gap this is $\min\{\rho, h\sqrt{\rho}/d\}$. For rarity bounded below, $\rho\omega_\mu(h) \le \Psi(h) \le \omega_\mu(h)$, by choosing $w \equiv \rho$ and by $w \le 1$ respectively. These comparisons have constants depending on the lower bound for ρ.

The inherited lower bound $R \gtrsim \Psi(B^{-1/2})$ (and its confidence analogue) is PROVED-ON-PAPER with the imported Lee–Valiant Corollary 19 and a bounded-likelihood-ratio decomposition as dependencies. The source requires simultaneous universal smallness of its prior, increment, gap, and relative increment; it is an expected-work, history-averaged statement. The continuation handles larger gaps separately by sign revelation and data processing. The fresh-reservoir reduction and its limits remain paper-level. The audit found no blocking gap in this argument; it is not an independently formalized minimax theorem. See `REVIEW_MATH.md` and `LITERATURE_AUDIT.md` for the exact inspected material.

Matching paper rates for $0 < \alpha < 2$ and the separate $\alpha > 2$ result were inherited, not proved for the first time by this run. For $\alpha > 2$, constants depend on α and diverge near two. The next section addresses the missing critical case but leaves an iterated logarithm. The heuristic $\rho\omega_\mu((\rho B)^{-1/2})$ remains DISPROVED: $\mu = \rho\delta_{\rho^2} + (1-\rho)\delta_{1/4}$ and $B = \rho^{-4}$ give true lower order ρ versus heuristic order ρ^2. No counterexample to Ψ itself was established here.

3 A critical-margin upper bound with a remaining iterated logarithm

PROVED-ON-PAPER. The result below is new work in this run, reviewed by a separate adversarial agent. It is not Lean-checked or externally reviewed; priority is unresolved. It does not settle the candidate minimax law. Natural logarithms are used throughout.

Assume the iid hidden-gap Bernoulli experiment of the question. Only requested coin observations are charged; arithmetic and independent randomness are free. For known $0 < a \le 1/2$, assume

$$\mu((0,x]) \le (x/a)^2 \quad (0 < x < a). \tag{1}$$

The individual gaps, signs, and unknown measurable function q remain hidden. Only the global condition $\int q\, d\mu \le \rho$ is imposed, with known $0 < \rho \le 1/4$. Set $L = 1+\log(1/\rho)$ and $H = 1+\log L$.

Theorem 3.1. *There is an observable single-coin estimator Y, depending only on a, ρ, such that uniformly over* (1) *and every admissible q,*

$$\mathbb{E}Y = \theta, \qquad \mathbb{E}Y^2 \le \frac{165}{56}\rho < 4\rho, \qquad \mathbb{E}T_Y \le W := 1350a^{-2}LH.$$

Thus, for every $B > 0$, an algorithm of uniform expected work at most B has worst-case RMS at most

$$\min\left\{\rho/2, \sqrt{5400\rho LH/(a^2B)}\right\}.$$

For $0 < \delta \le 1/4$, let k be the least odd integer at least $8\log(1/\delta)$. A confidence radius

$$\min\{\rho/2, 4\sqrt{\rho Wk/B}\}$$

is attainable with the same uniform expected-work budget. These statements give no deterministic work cap or high-probability work bound.

3.1 An entirely observable finite classifier

For $j \ge 0$, put

$$\varepsilon_j = 16^{-(j+1)}, \quad N_j = \left\lceil (a^2\rho\varepsilon_j)^{-1} \right\rceil, \quad K_j = \left\lceil \log_2(a\sqrt{N_j}) \right\rceil.$$

For $0 \le k \le K_j$, let $r_k = a2^{-(k+1)}$. If H_n is the number of observed heads after n flips of the current coin, define

$$L^+_{k,n} = (1+2r_k)^{H_n}(1-2r_k)^{n-H_n}, \qquad L^-_{k,n} = (1-2r_k)^{H_n}(1+2r_k)^{n-H_n}.$$

At each $n < N_j$, output zero and stop if some $L^-_{k,n}$ reaches $(K_j+1)/\varepsilon_j$, or output one and stop if some $L^+_{k,n}$ reaches $(K_j+1)/(\rho\varepsilon_j)$. Give zero priority in a simultaneous crossing. If neither decision has occurred before N_j, output $\mathbf{1}_{\{H_{N_j} > N_j/2\}}$. Call the output Z_j and work T_j. Only finitely many likelihoods are checked at any time.

Under the wrong sign the corresponding likelihood is a nonnegative supermartingale with expected multiplier $1 - 4r_k d \le 1$. Applying the bounded-horizon crossing inequality to each of the K_j+1 likelihoods gives wrong-crossing probability at most ε_j on positives and $\rho\varepsilon_j$ on negatives. The wrong forced-majority event is a subset of the full-tape majority-error event, whose conditional probability is at most $\exp(-2N_jd^2)$ by Hoeffding. No independence between stopping and this event is used. Stochastic dominance from (1) yields

$$\mathbb{E}_\mu e^{-2ND^2} \le \frac{1-e^{-2Na^2}}{2Na^2} \le \frac{1}{2Na^2}.$$

Consequently, for every admissible q,

$$\mathbb{P}(Z_j \neq S) \leq \tfrac{5}{2}\rho\varepsilon_j, \qquad \mathbb{P}(Z_j = 1, S = 0) \leq \tfrac{3}{2}\rho\varepsilon_j. \tag{2}$$

The $\theta\varepsilon_j$ positive error was bounded by $\rho\varepsilon_j$; the assumption was never strengthened to $q(d) \leq \rho$. For completeness, the concentration input has a direct Bernoulli proof: the centered log moment-generating function $\phi(\lambda)$ has $\phi(0) = \phi'(0) = 0$ and $\phi''(\lambda) \leq 1/4$, because the tilted Bernoulli variance is at most $1/4$. Hence $\phi(\lambda) \leq \lambda^2/8$. Multiply these bounds for independent observations and apply Markov's inequality with $\lambda = 4d$ (or $-4d$) to obtain e^{-2Nd^2}. This argument also applies to the independent failure indicators used below.

3.2 Conditional work and global-rarity rearrangement

Write $x = \min(d, a)$ and

$$A_s = 1 + \log(K_j + 1) + \log(1/\varepsilon_j) + s\log(1/\rho), \quad s = 0, 1.$$

If $x \geq N_j^{-1/2}$, choose $k = \lceil \log_2(a/x) \rceil \leq K_j$. Then $x/4 \leq r_k \leq x/2$, $r_k \leq 1/4$, and the correct-sign log-likelihood has iid increments with mean

$$g = \tfrac{1}{2}\log(1 - 4r_k^2) + d\log\frac{1 + 2r_k}{1 - 2r_k} \geq -\tfrac{8}{3}r_k^2 + 4dr_k \geq \tfrac{2}{3}x^2.$$

Here $\log(1 - z) \geq -z/(1 - z)$ and $\log((1 + u)/(1 - u)) \geq 2u$ suffice. Every upward increment is less than one. For the crossing time σ of the correct threshold, the bounded stopped-sum identity at $\sigma \wedge n$ gives $g\mathbb{E}(\sigma \wedge n) \leq A_s$. Monotone convergence therefore gives $\mathbb{E}\sigma \leq (3/2)A_s/x^2$. The capped classifier work is at most $\min(N_j, \sigma)$, including when the crossing occurs exactly at the cap. If $x < N_j^{-1/2}$ the same conclusion below holds trivially because $A_s \geq 1$:

$$\mathbb{E}(T_j \mid d, s) \leq \min\{N_j, (3/2)A_s/x^2\}. \tag{3}$$

Set $c = 3/2$. For the negative-sign contribution stochastic dominance gives

$$\mathbb{E}\min\{N, cA_0/x^2\} \leq cA_0a^{-2}[1 + \log_+(Na^2)].$$

For positives, layer cake and $\mathbb{E}[q\mathbf{1}_E] \leq \min\{\rho, \mu(E)\}$ give

$$\begin{aligned}\mathbb{E}\big[q(D)\min\{N, cA_1/x^2\}\big] &\leq cA_1a^{-2}[1 + \log_+(Na^2\rho/(cA_1))] \\ &\leq cA_1a^{-2}[1 + \log_+(Na^2\rho)].\end{aligned}$$

Indeed the sharp upper is the integral of $\min\{N, cA_1/(a^2u)\}$ over $0 < u < \rho$. If $Na^2\rho \leq cA_1$ this is $N\rho \leq cA_1/a^2$; otherwise it is exactly $(cA_1/a^2)[1 + \log(Na^2\rho/(cA_1))]$. This proof also covers envelope laws with atoms, through layer cake rather than a quantile-density assumption. Bounding the negative part by the full unweighted integral and adding the positive part bounds $\mathbb{E}T_j$.

The following loose universal bounds follow directly from the definitions:

$$\begin{aligned} N_ja^2 \leq 2/(\rho\varepsilon_j), \qquad & K_j + 1 \leq 5(L + j + 1), \\ A_0 \leq 7H(j + 1), \qquad & A_1 \leq 7H(j + 1) + L, \\ 1 + \log(N_ja^2) \leq 5L(j + 1), \qquad & 1 + \log(N_ja^2\rho) \leq 5(j + 1). \end{aligned}$$

For example, use $\log(L + j + 1) \leq \log L + j + 1$ and $4\log 2 < 3$ in the bound on A_0. Since $L, H \geq 1$, substitution proves

$$\mathbb{E}T_j \leq (225/2)a^{-2}LH(j + 1)^2. \tag{4}$$

3.3 Randomized telescoping and exact expected-budget aggregation

Independently draw J with $p_j = 2^{-(j+1)}$, then draw a fresh coin. For $J = 0$, set $\Delta_0 = Z_0$. For $J = j \geq 1$, run Z_j and Z_{j-1} on two successive disjoint observation blocks of that *same* coin and set $\Delta_j = Z_j - Z_{j-1}$. Return $Y = \Delta_J / p_J$.

Conditional on (D, S), the observations after the first bounded stopping time again form an iid Bernoulli tape with the original parameter. To verify this directly, condition on each possible first stopping time n and each observed length-n prefix. Its event is prefix-measurable and the unused iid coordinates have their original law. Summing the finitely many prefix events proves the suffix assertion. Thus the second block has exactly the classifier marginal used below; its work can be added in expectation. The two outputs share their latent sign, which is essential to the disagreement bound.

By (2),

$$\mathbb{E}\Delta_0^2 \leq \tfrac{35}{32}\rho, \qquad \mathbb{E}\Delta_j^2 \leq \tfrac{85}{32}\rho 16^{-j} \quad (j \geq 1).$$

For the latter, different decisions imply that at least one is wrong about the shared sign. Hence Tonelli gives

$$\mathbb{E}Y^2 \leq \frac{35}{16}\rho + \frac{85}{16}\rho \sum_{j \geq 1} 8^{-j} = \frac{165}{56}\rho < 4\rho.$$

The same bounds give $\sum_j \mathbb{E}|\Delta_j| < \infty$; signed expectations may therefore be interchanged and telescoped, giving $\mathbb{E}Y = \lim_j \mathbb{E}Z_j = \theta$. This proves integrability before using the signed telescope. The expected work of level j is at most twice (4), and $\sum_j p_j (j+1)^2 = 6$. Thus $\mathbb{E}T_Y \leq 1350 a^{-2} LH$. Every call terminates almost surely, and the resulting expected work is finite.

For budget B, set $\lambda = B/W$, independently draw $M \sim \mathrm{Poisson}(\lambda)$, and average the independent fresh-coin outputs as $\lambda^{-1} \sum_{i=1}^{M} Y_i$. Conditioning on M proves expectation θ, variance $\mathbb{E}Y^2/\lambda \leq 4\rho W/B$, and expected work at most B. This remains valid for $\lambda < 1$. Project onto $[0, \rho]$, which improves squared error without preserving unbiasedness. If the stated RMS bound exceeds $\rho/2$, choose the zero-work constant $\rho/2$ instead. This proves the RMS assertion.

For confidence, run k independent copies with budgets B/k and take their median. Each has failure probability at most $1/4$ at radius $4\sqrt{\rho W k/B}$ by Chebyshev. Hoeffding's inequality for the independent failure indicators bounds median failure by $e^{-k/8} \leq \delta$. If $\rho/2$ is the smaller displayed radius, explicitly choose the constant estimator instead. Total expected work is at most B.

3.4 Exact critical candidate and the unresolved gap

For the exact law $F(d) = (d/a)^2$, direct integration yields

$$\rho = (t/a)^2[1 + 2\log(a/t)], \qquad \Psi(h) = \begin{cases} [2ht(1 + \log(a/t)) - h^2]/a^2, & 0 < h < t, \\ \rho, & h \geq t. \end{cases}$$

Set $u = 1 + 2\log(a/t)$. Then $\rho = ue^{1-u}$ and $L = u - \log u$, whence $L \leq u \leq 2L$. For $h < t$, $\Psi(h)/h = (t/a^2)(u + 1 - h/t)$ lies between tu/a^2 and $t(u+1)/a^2$. Since $tu/a^2 = \sqrt{\rho u}/a$, including the transition at $h = t$ proves

$$\Psi(h) \asymp \min\{\rho, h\sqrt{\rho L}/a\}$$

with universal constants, jointly in ρ, h, a. Conditional on the audited arbitrary-law lower theorem, Theorem 3.1 therefore leaves a factor at most $\sqrt{1 + \log L}$. It improves the critical upper bracket but does not remove this factor, prove it necessary, or establish arbitrary-law optimality. At fixed positive ρ, the construction gives $B^{-1/2}$ RMS; this fixed-rarity statement must not be confused with a uniform rare-event matching theorem.

4 A finite, arbitrary-law upper bracket

PROVED-ON-PAPER; established allocation principle. The classifier analysis also gives a constructive upper bracket for every law on $(0, 1/2]$, including atoms, singular laws, and infinite inverse moments. No comparison of this bracket to Ψ is asserted.

Set $a = 1/2$. Choose finitely many integer caps $N_j \geq 4$ and tolerances $0 < \varepsilon_j \leq 1$, $0 \leq j \leq m$, and use the preceding classifiers. Let

$$\beta_j = \mathbb{E}_\mu e^{-2N_j D^2}, \quad e_j = 2\rho\varepsilon_j + \beta_j, \quad A_{j,s} = 1 + \log(K_j + 1) + \log(1/\varepsilon_j) + s\log(1/\rho).$$

For a bounded nonnegative measurable function f, define

$$Q_\rho(f) = \int_0^\infty \min\{\rho, \mu(f > v)\}\, dv.$$

Layer cake gives $\int qf\, d\mu \leq Q_\rho(f)$ for every feasible q. Equality follows by choosing the largest ρ mass of f, with a fractional value of q on a cutoff atom. Thus this formula does not require a nonatomic law. Put

$$f_{j,s}(d) = \min\{N_j, (3/2)A_{j,s}/d^2\}, \qquad c_j = \mathbb{E}f_{j,0} + Q_\rho(f_{j,1}).$$

The preceding conditional-work proof gives a uniform bound c_j; all these integrals are finite because the functions are bounded. Let

$$v_0 = \rho + \rho\varepsilon_0 + \beta_0, \quad C_0 = c_0, \qquad v_j = e_j + e_{j-1}, \quad C_j = c_j + c_{j-1} \quad (j \geq 1).$$

Use independent copies of the same-coin increments, independently across levels. With $S_m = \sum_{j=0}^m \sqrt{v_j C_j}$, independently take Poisson numbers of copies at means $\lambda_j = B\sqrt{v_j/C_j}/S_m$. Their sum, with level j divided by λ_j, has expectation $\mathbb{E}Z_m$, variance at most S_m^2/B, and expected work at most B. Therefore

$$R_{\mu,\rho}(B) \leq \min\{\rho/2, \sqrt{e_m^2 + S_m^2/B}\}.$$

All constants here are positive. The RMS bound uses the exact bias–variance identity, and clipping onto $[0, \rho]$ cannot increase risk.

The known law can determine these finitely many design constants, or certified upper bounds on them, before sampling. This gives no access to the magnitude of a sampled coin. For an abstract probability measure this is existence at the fixed known law; uniform computability from every representation of a measure is not claimed. Floating-point quadrature alone supplies numerical design evidence, not certified risk constants. The Poisson allocation and randomized telescoping principles are established methods; no novelty is claimed for them.

5 A reusable bounded-channel upper principle

PROVED-ON-PAPER; no lower transfer. The critical theorem and arbitrary-law upper bracket hold with the same constants if each hidden state produces conditionally iid observations $X \in [0, 1]$ of mean $1/2 + (2S - 1)D$. Charge one unit per observation; the algorithm receives no hidden magnitude, sign, or selected-state access. Replace the reference processes by

$$M_{k,n}^{\pm} = \prod_{i=1}^n [1 \pm 2r_k(2X_i - 1)]$$

and keep the same thresholds, caps, and empirical-mean decision. The factors are positive. Under the wrong sign their conditional mean is $1 - 4r_k d \le 1$. For the correct sign, concavity of the logarithm gives

$$\log(1 + 2r(2X - 1)) \ge X \log(1 + 2r) + (1 - X) \log(1 - 2r).$$

Its expected log increment is therefore at least the Bernoulli drift used above; upward increments remain below one. For the concentration step the second derivative of a centered log moment-generating function is the variance under a tilted law on $[0, 1]$, at most $1/4$. Hence the previous mgf proof applies. All conditional work, global-rarity integration, and telescoping arguments carry over unchanged. This rule directly uses the real observations.

For a concrete derived synthetic example, take, conditional on d, s,

$$X \sim \mathrm{Unif}[1/2 + (2s - 1)d - 1/4,\ 1/2 + (2s - 1)d + 1/4], \quad 0 < d \le a \le 1/4.$$

With independent noises on revisits this is a valid bounded channel with variance $1/48$. The quadratic-margin upper theorem applies, charging one real draw per step. No minimax equality or empirical application is asserted.

An exact information-discarding reduction is also available: using an independent $U \sim \mathrm{Unif}[0, 1]$, the variable $\mathbf{1}_{\{U \le X\}}$ is a Bernoulli observation with the required conditional mean. One real observation is charged per converted bit and the target is unchanged. This transfers Bernoulli upper algorithms; it does not transfer a Bernoulli lower bound to every channel. If instead X is deterministically its conditional mean, one observation reveals S, and n fresh states yield variance at most ρ/n without an inverse-gap factor. Nothing here extends a bounded-observation theorem to unbounded observations.

Extra cross-state structure also changes the question. As an explicit different experiment, suppose the outer state $Z \sim \mathrm{Unif}[-1, 1]$ is observed, $\beta \in [-1/2, 1/2]$ is unknown, and

$$X = \tfrac{1}{2} + (Z - \beta)/8 + \xi, \qquad \xi \sim \mathrm{Unif}[-1/16, 1/16].$$

Independent pairs have $X \in [0, 1]$, and the positive-mean fraction is $\theta = (1 - \beta)/2$. The average of $Z - 8(X - 1/2)$ is unbiased for β with variance $1/(12n)$; the corresponding estimate of θ has variance $1/(48n)$. Charging one unit per outer draw and one per inner observation gives work $2n$. The margin near zero is linear, yet this common regression parameter permits $B^{-1/2}$ RMS without a margin logarithm. Observed states and cross-state structure are explicit extra information, so this calculation does not contradict the hidden unstructured reservoir lower theorem.

6 A necessary correction to the countable-reservoir bridge

Evidence status. The next two propositions are PROVED-ON-PAPER in this run. Their stochastic statements are not Lean-checked. Exact rational finite-prefix checks are reported separately and do not certify the infinite-horizon calculation. The finite-coin Tier 3 theorem is unchanged and is not contradicted.

Proposition 6.1 (One-law finite work does not transfer termination)**.** *Consider fresh iid coins with deterministic gap $D = 1/2$. Under P their positive sign probability is $5/8$, and under Q it is $3/8$. Query each coin once. If X_i is its observed bit, stop at*

$$T = \inf\{n \ge 1 :\ \textstyle\sum_{i=1}^{n} (2X_i - 1) = 1\}.$$

Then $P(T < \infty) = 1$, $\mathbb{E}_P T = 4$, whereas $Q(T < \infty) = 3/5$. In particular $Q(T = \infty) = 2/5$.

Proof. For iid bits of parameter p, put $q = 1 - p$. A first hit at time $2k + 1$ has $k + 1$ upward and k downward steps, and the first $2k$ partial sums never exceed zero. The number of such paths is the Catalan number $C_k = (k+1)^{-1}\binom{2k}{k}$. This follows by reflecting at the first forbidden positive step: among paths with k up and k down steps, the forbidden paths are counted by $\binom{2k}{k+1}$, whose subtraction gives C_k. Thus

$$\Pr_p(T = 2k + 1) = C_k p^{k+1} q^k.$$

The generating series $C(z) = \sum_{k\geq 0} C_k z^k$ satisfies $C(z) = 1 + zC(z)^2$: split a nonempty Catalan path at its first return to zero. The solution continuous at zero with $C(0) = 1$ is $C(z) = (1-\sqrt{1-4z})/(2z)$ for $0 \leq z < 1/4$. Consequently the probability generating function, excluding the mass at infinity, is

$$G_p(s) = \mathbb{E}_p[s^T; T < \infty] = psC(pqs^2) = \frac{1 - \sqrt{1 - 4pqs^2}}{2qs}, \qquad 0 < s \leq 1.$$

For $p \neq 1/2$ all series converge in a neighborhood of $s = 1$. It follows that $G_p(1) = 1$ when $p > 1/2$, and $G_p(1) = p/q$ when $p < 1/2$. For $p > 1/2$, differentiation gives $G_p'(1) = 1/(2p - 1)$. Equivalently differentiate $G_p(s) = s[p + qG_p(s)^2]$ at $s = 1$; nonnegative-series differentiation computes $\mathbb{E}_p T$ because $G_p(1) = 1$. Substituting $p = 5/8$ and $p = 3/8$ gives the claims. □

The example lies exactly in the symmetric prior family $1/2 \pm \eta$ with $\eta = 1/8$. It also has an interior-gap variant: choose $D = 1/4$, retain the same sign priors, and use only the first observation of each fresh coin. The observed bits then have parameters $9/16$ and $7/16$; the same calculation gives $\mathbb{E}_P T = 8$ and $Q(T = \infty) = 2/9$. Thus deterministic coin endpoints are not responsible for the failure. For a fixed finite coin family, the two latent-prior laws have a globally bounded likelihood ratio; for an unbounded reservoir that finite-family comparison is unavailable.

This example still has finite full-transcript relative entropy. On every finite stopped path the upward step count exceeds the downward count by one, so the likelihood ratio dP/dQ is $5/3$. On infinite paths it is zero, since P has no such mass. Therefore $\mathrm{KL}(P, Q) = \log(5/3)$. It is consistent with the proposed full-transcript bound at $h = 1/2$: convexity of $1/x$ gives $\log(5/3) \leq 8/15 < 128/225 = (128/225)h^2\mathbb{E}_P T$. Thus full KL can extend while the two-law stopping transfer cannot.

Proposition 6.2 (Finite-prefix information implies terminal testing)**.** *Let P, Q describe a common policy with independent seed included in increasing observable prefix sigma fields $\mathcal{F}_n$. Let T be its stopping time. Assume $T < \infty$ almost surely under each law and $\mathbb{E}_P T \leq B < \infty$. Suppose, for a constant $k \geq 0$, every finite-prefix transcript satisfies*

$$\mathrm{KL}(P|_{\mathcal{F}_n}, Q|_{\mathcal{F}_n}) \leq k\mathbb{E}_P(T \wedge n).$$

If a terminal test $\phi \in \{0, 1\}$ has $P(\phi = 1) \geq 1 - \delta$ and $Q(\phi = 1) \leq \delta$, with $0 < \delta < 1/2$, then

$$\mathrm{kl}(1 - \delta, \delta) \leq kB.$$

Proof. The event $A_n = \{T \leq n, \phi = 1\}$ belongs to $\mathcal{F}_n$ because the test is measurable at stopping. Let $p_n = P(A_n)$ and $q_n = Q(A_n)$. Then $p_n \geq 1 - \delta - P(T > n) \geq 1 - \delta - B/n$, while $q_n \leq \delta$. For all sufficiently large n, $1 - \delta - B/n > \delta$. Data processing for the binary indicator of A_n, and monotonicity of binary relative entropy in its first argument above its second, and in its second argument below its first, give

$$\mathrm{kl}(1 - \delta - B/n, \delta) \leq \mathrm{kl}(p_n, q_n) \leq \mathrm{KL}(P|_{\mathcal{F}_n}, Q|_{\mathcal{F}_n}) \leq kB.$$

Continuity at $(1-\delta,\delta)$ proves the result by $n\to\infty$. The case $q_n=0<p_n$ gives infinite binary relative entropy and cannot occur under a finite information bound. No transfer of termination from P to Q has been used. Separate finite expected-work bounds under the two laws are a sufficient condition for the stated two termination assumptions. □

Canonical numbering and the remaining formal obligation. After n charged observations, at most n coin identities have been observed. Assign indices $1,\dots,n$ in order of first observation and retain a dictionary for revisits. Every unobserved iid latent draw can be deferred until its first observation: prior decisions involving its label contain no information about that draw. This recursive construction preserves the observed finite-prefix law and the number of charged observations. It requires that each action completes after finite computation; infinitely many cost-free operations do not constitute an observation oracle. For the symmetric information bound, condition on the first n iid magnitudes and the independent seed, apply the deterministic-gap finite-prefix bound, and integrate. Magnitudes may be appended for this upper bound on information and then removed by data processing; the policy itself does not observe them. For the rare-prior bound, the appropriate single-coin inequality and bounded-ratio hybrid decomposition must instead be used.

This is a paper-level reservoir realization and reduction, not an existing Lean theorem. The existing finite-family declaration cannot be instantiated with a different coin count at each horizon and thereby produce a single already formalized infinite-reservoir process. One must formalize its joint law and prove consistent realization of every prefix. The terminal-testing proposition then closes the limit once those prefix bounds and the policy's admissibility are available. The minimax experiment's uniform expected budget gives finite work under both alternatives, so the counterexample does not invalidate the paper minimax lower bound.

Reusable scope and boundary. The terminal-testing bridge works for any observation channel and any positive cost accounting once its finite-prefix KL bound has been proved. With unequal costs, a proved bound of the form $\mathrm{KL}_n\le c\mathbb{E}_P\sum_{j\le T\wedge n}a_j$ gives the corresponding testing bound by the same limiting argument, provided stopping is established separately (a positive lower bound on a_j is one sufficient condition). There is no asserted universal Bernoulli scaling for other channels. Cross-state information can invalidate the iid coin likelihood factorization even though this abstract testing step still holds; a shared unknown parameter learned at one state must be included in a different finite-prefix information calculation.

7 A qualified guarantee when rarity is unknown

Evidence status. PROVED-ON-PAPER in this run, not Lean-checked and not claimed original. This elementary construction supplies a sufficient additive floor, not a matching adaptation penalty.

Proposition 7.1 (Known fixed gap, unknown rarity)**.** *Let $\mu=\delta_d$, $0<d<1/2$, be known. For budget $B\ge 1$, let m be the least positive odd integer at least $\log(2B)/(2d^2)$. If $B\ge 2m$, there is one estimator independent of ρ such that, for every $0\le\theta\le 1$,*

$$\mathbb{E}_\theta T\le B,\qquad \mathbb{E}_\theta(\widehat{\theta}-\theta)^2\le\frac{8m\theta}{B}+\frac{4m}{B^2}.$$

Its work actually has a deterministic cap. For $B<2m$, the constant zero has error θ. These two cases specify the algorithm without knowing θ or a rarity bound.

Proof. Set $n = \lfloor B/m \rfloor$, and query n fresh coins exactly m times each. Let Z_i be their strict-majority decisions. Put

$$f = \Pr\{\mathrm{Bin}(m, 1/2 - d) > m/2\}.$$

This number is computable from d, m, with no access to the hidden sign. Oddness gives positive-sign success probability $1 - f$, so Z_i are iid Bernoulli with mean $p = f + (1 - 2f)\theta$. Return $(\overline{Z} - f)/(1 - 2f)$, clipped to $[0, 1]$. Hoeffding's inequality gives $f \le e^{-2md^2} \le 1/(2B) \le 1/4$. The raw estimator is unbiased and has variance

$$\frac{p(1-p)}{n(1-2f)^2} \le \frac{4(\theta + f)}{n} \le \frac{8m\theta}{B} + \frac{4m}{B^2},$$

using $n \ge B/(2m)$. Clipping reduces squared error. The cost is $nm \le B$; termination is deterministic. This proof requires no random-budget argument. □

Thus for a known fixed interior gap this one estimator has, uniformly over $\theta \le \rho$, error of order at most $\sqrt{\rho \log(2B)/(d^2 B)} + \sqrt{\log(2B)}/(dB)$ in the nontrivial regime. The logarithmic multiplicative loss and additive floor are *sufficient*; the qualitative unknown-rarity impossibility does not establish either as necessary. No claim of an optimal adaptation rate follows.

Unlike a distribution-dependent oracle radius, this construction also provides a genuinely computable, although not rarity-sharp, confidence interval. For $B \ge 2m$, set

$$r = \frac{\sqrt{\log(2/\delta)/(2n)}}{1 - 2f}$$

and intersect $[(\overline{Z} - f)/(1 - 2f) - r, (\overline{Z} - f)/(1 - 2f) + r]$ with $[0, 1]$. The Bernoulli Hoeffding inequality gives coverage at least $1 - \delta$. For smaller budgets return $[0, 1]$. Exact arithmetic is the mathematical contract; a floating approximation to f must use certified bounds or add its numerical error to the interval. This interval was not benchmarked here.

At the deterministic endpoint $d = 1/2$, one flip reveals the sign, and $n = \lfloor B \rfloor$ flips give the ordinary binomial estimator with variance $\theta(1 - \theta)/n$. It needs no rarity input and vanishes surely at $\theta = 0$. The inherited impossibility explicitly excludes this endpoint.

Other adaptation questions. The critical-margin theorem above already handles unknown individual μ inside a supplied quadratic envelope, with known (a, ρ); the inherited $\alpha > 2$ construction handles its own envelope class. Neither proves distribution-blind attainment of Ψ for arbitrary unknown μ. When both μ and ρ are unknown, the inherited v4 unrestricted distribution-blind bound still applies, with its population-dependent modulus; it is not an observable confidence radius or a rarity-optimal guarantee. Matching necessity for qualified adaptation remains open here.

8 What the existing nested-simulation lower bound settles

PRIOR-ART / UNRESOLVED-PRIORITY. The detailed audit inspected primary theorem statements and relevant proofs, not just abstracts. The v4 observed-state construction satisfies GHA's Assumptions 2.1 and 2.4 literally. Fixing $a_0 = 1/4$, its normalized gap has density $[2a_0(1 + z^2)^{3/2}]^{-1} \le 2$, and every conditional standardized kth absolute moment is at most $3^{k/2}$. Both constants are independent of accuracy and the number of cells. The fixed admissible class is the union of all finite cell constructions with arbitrary finite sign arrays. Its hard member may depend on accuracy. A fixed finite list of cells is not the claimed asymptotic hard instance.

Thus the inherited lower theorem concerns an unstructured observed-state class meeting the stated adaptive-MLMC assumptions, even with selected-state queries. A fixed per-outer-draw cost changes its upper accounting by a constant factor. This does not establish a lower bound for the specific known Gaussian GHA simulator, conditional-Gaussian-only classes, smooth cross-state regression classes, or rare-event subclasses. Broadie–Du–Moallemi's conditional-normal premise and Gordy–Juneja's joint-density premise exclude the literal Bernoulli hard family. Smooth nested simulation and nested kernel quadrature have additional cross-state or outer-function regularity.

The audit also located Blum–Hu's earlier adaptive good-arm fraction problem. Inverse-binomial calibration and randomized telescoping have explicit predecessors in Mendo/Haldane, McLeish, Rhee–Glynn, and Vihola. The inherited audit's assertion that Lee–Valiant used a different Hellinger normalization was wrong: Definition 14 uses the same $1-$affinity convention. This correction is recorded separately without altering history. No priority claim follows from this bounded search, and specialist correspondence remains unsent.

9 Practical evidence on a fixed, disclosed suite

NUMERICAL; no superiority claim. All four historical direct-real versus tuned-GHA cells were resimulated with their original seeds and 500 replications each. All 2,000 raw outputs and work counts matched exactly. At budget 16,384 the clipped RMS values remain 0.02990 for tuned GHA and 0.03910 for the historical candidate; at 65,536 they remain 0.02012 and 0.02185. The earlier rejection is preserved.

The new suite froze implementation hashes, seven simulators, two budgets, 128 replications per cell, metrics, seeds, and exclusion rules before running 42 cells. The candidate changes the positive sample-variance threshold of a two-level direct-real randomized telescope by $\sqrt{1+\log(1/\rho)}$. This is a heuristic, not an implementation of the proved likelihood-test theorem. It is compared to its symmetric ablation and the GHA Algorithm 1 inner-sampling component in a single-level, independently calibrated driver. The latter is explicitly an altered driver, not the original tuned full comparator; favorable results against it do not establish superiority to GHA. All methods receive the same original real observations, with no Bernoulli conversion. Every pilot and production observation is charged; fresh outer counts and alternative accounting charging them are also recorded.

The models are the published GHA simulator, a rarer version, a fixed Bernoulli gap, two separated gap scales, near-threshold negative coins, an independently specified uniform-mean Gaussian simulator, and deterministic endpoint coins. Their ground truths are analytic. Synthetic models are labelled synthetic. All probabilities are nonzero. Zero, half, and supplied-rarity-half constants are reported; some latter constants coincide with a particular simulator's truth, a coincidence that confers no uniform guarantee.

The candidate is rejected as a general practical improvement: results are mixed, and the near-negative model has RMS around 0.43 even at the larger budget while the true probability is 0.05. No matched-accuracy speedup or break-even workload is established. New fitting and tuning costs are zero; verification, baseline replay, and historical development costs are separately recorded. Final clipping improves squared loss but changes bias and leaves cost unchanged. There is no confidence procedure for this heuristic, so coverage is marked not applicable. Bootstrap intervals are descriptive only.

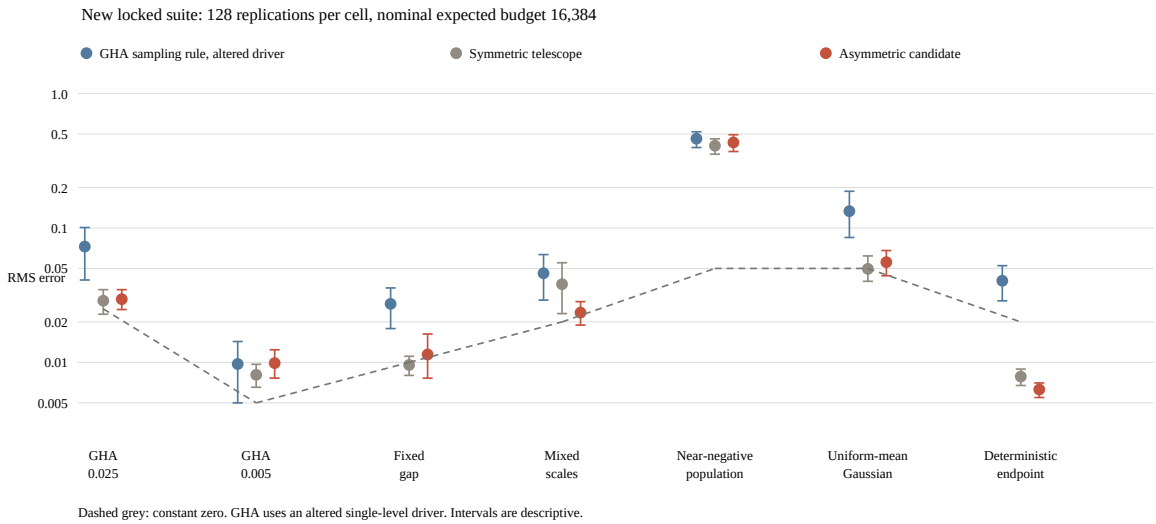


Figure 1: New held-out numerical evidence, budget 16,384. Error bars are descriptive bootstrap intervals. The grey line is the zero-estimator error; the GHA curve uses an altered single-level driver. No uniform gain follows.

10 Reproducibility and remaining obligations

The package contains a claim ledger, dependency graph, source audit, algorithm contracts, fixed test configuration, raw samples, exact finite diagnostics, current-source formal audit records, and a failed-approach log. The initial fourteen numerical/integrity gates passed on a byte snapshot rather than overwriting historical results. Kernel coverage and any concurrent-source limitations are specified in `FORMAL_COVERAGE.md`; a build verifies only its audited statements. New statistics are not certified by those builds.

The central unresolved obligation is a universal Ψ upper theorem or a rigorous counterexample. Even for the quadratic power law, necessity or removal of $1 + \log L$ is open. Unknown-law adaptation outside specified envelopes, sharp unknown-rarity penalties, and a formal countable-reservoir realization with all information bounds are not completed here. This is a finite research checkpoint, not a promise of background continuation.

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